

A Concise Proof of the L_0 Dichotomy

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Abstract

Carroy, Miller, Schritterser, and Vidnyánszky established the L_0 dichotomy: there is a Borel graph of Borel chromatic number three that admits a continuous homomorphism to every analytic graph of Borel chromatic number at least three. Their proof relies on a transfinite analysis of terminal approximations over a decreasing ω_1 -sequence of analytic sets.

This brief note offers a new, substantially shorter proof of this result by adapting the graph-theoretic framework recently introduced by Bernshteyn for the G_0 dichotomy. The central device is a σ -ideal of small sets of homomorphisms from finite path approximations into the target graph, where smallness is witnessed by a bounded odd-walk condition on vertex projections. The key lemma that largeness is preserved under the doubling operation is established via the First Reflection Theorem, replacing the original transfinite construction with a single Borel reflection argument. The continuous homomorphism from the canonical graph $\mathbb{L}_c$ into the target is then obtained as a limit of shrinking families of copies, in direct analogy with Bernshteyn's proof for G_0 .

1 Introduction

Throughout this entire work, G denotes an analytic graph on a Polish space $V(G) = X$. That is, X is a separable completely metrizable topological space, and G is a symmetric and irreflexive relation on X which, as a subspace of $X \times X$, is the continuous image of a Polish space.

Denote the *Borel chromatic number* of G by $\chi_B(G)$, that is, $\chi_B(G)$ is the least number of colours k needed to find a Borel mapping $c: X \rightarrow k$ in such a way that adjacent vertices in G are given different colours. We employ the symbol $H \rightarrow_c G$ (respectively, $H \rightarrow_B G$) to abbreviate the fact that there is a continuous (respectively Borel) homomorphism from the analytic graph H to the analytic graph G . It is straightforward to show the following.

Fact 1.1. *If $H \rightarrow_c G$ or, more generally if $H \rightarrow_B G$, then $\chi_B(H) \leq \chi_B(G)$.*

Using a standard greedy algorithm argument, it is clear that any finite graph G of maximum degree Δ satisfies $\chi(G) \leq \Delta + 1$. Interestingly, a similar fact holds for Borel graphs of uniformly bounded degree.

Theorem 1.2 (Proposition 4.6 in [KST99]). *If G is a Borel graph on a Polish space all of whose degrees are bounded by the natural number k , then $\chi_B(G) \leq k + 1$.*

The following notion, taken from [Kec95, Theorem 35.10, p. 285], is a well-known fact about analytic sets.

Definition 1.3. A collection of subsets Φ of a Polish space X is said to be Π_1^1 on Σ_1^1 if for any Polish Y and any Σ_1^1 set $A \subseteq Y \times X$, the set $A_\Phi := \{y \in Y : A_y := \{x \in X : (y, x) \in A\} \in \Phi\}$ is Π_1^1 .

Lemma 1.4 (First Reflection Theorem). *If Φ is as in the preceding definition, then for any Σ_1^1 set $A \in \Phi$, there is a Borel set $B \subseteq X$ such that $A \subseteq B$ and $B \in \Phi$.*

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2 Finite path approximations

Fix the parameter $c \in \omega^\omega$ and a path of length ω on the vertex set $\{p_n\}_{n < \omega}$. That is, the edge set $\{(p_n, p_{n+1}) : n < \omega\}$. Naturally, one could assume that $p_n = n$, but the labeling helps distinguish these vertices from other indices.

Recall that if s and t are finite strings (of, say, natural numbers), then $s \frown t$ is the concatenation of s and t . Similarly, t^n is the repeated concatenation of t with itself, and (i) denotes the string of single value i . Next, recursively construct a family of graphs $\{L_n^c : n < \omega\}$ such that for all $n < \omega$: L_n^c is a finite path; if $n > 0$, the endpoints of L_n^c are $e_i^n := (p_0) \frown (0)^{n-1} \frown (i)$, for $i < 2$; and

$$\text{Every vertex of } L_n^c \text{ is of the form } (p_k) \frown t \text{ for some } k \text{ and } t \in 2^{\leq n}. \quad (1)$$

Unless $t = \emptyset$, call such vertex a *non-path vertex*, and call the vertices of the form (p_k) the *path vertices* of L_n^c .

Start with L_0^c as the graph consisting of a single vertex p_0 and no edges, and set $e_0^0 = e_1^0 = p_0$. Now suppose that L_n^c has been constructed. L_{n+1}^c will be a path obtained from joining two copies of L_n^c by a path of $c(n) + 2$ edges (with $c(n) + 1$ new path vertices), but one must update the vertex labels in order to distinguish the two copies. One accomplishes this by simply appending 0 to each vertex of the first copy, and 1 to the other. Formally, start by adding the vertex $v \frown (i)$ for every $i < 2$ and $v \in L_n^c$. Adding the path

$$((e_1^n) \frown (0), p_0, p_1, \dots, p_{c(n)}, (e_1^n) \frown (1))$$

$$e_i^{n+1} = (e_0^n)^\frown(i) = (p_0)^\frown(0)^n \frown(i),$$

Figure 1 illustrates the construction of a 2D lattice of L_2 copies. The lattice is organized into three horizontal rows, each representing a different L_2 copy. The nodes are colored based on their level: blue for L_0 copy, green for level-1 join, orange for level-2 join, and red for level-3 join. The nodes are connected by horizontal and vertical lines. Blue arrows indicate the direction of the lattice construction. The vertical axis is labeled $\text{join } c(2)=5$.

The lattice structure is as follows:

- Top Row (L_2 copy 0):** Nodes are (0,0), (1,0), (2,0), and (3,0). The nodes (0,0), (1,0), and (2,0) are blue (L_0 copy), and (3,0) is orange (level-2 join). The nodes are connected by horizontal lines. Blue arrows point from (0,0) to (1,0) and from (2,0) to (1,0).
- Middle Row (L_2 copy 1):** Nodes are (0,1), (1,1), (2,1), and (3,1). The nodes (0,1), (1,1), and (2,1) are blue (L_0 copy), and (3,1) is orange (level-2 join). The nodes are connected by horizontal lines. Blue arrows point from (0,1) to (1,1) and from (2,1) to (1,1).
- Bottom Row (L_2 copy 2):** Nodes are (0,2), (1,2), (2,2), and (3,2). The nodes (0,2), (1,2), and (2,2) are blue (L_0 copy), and (3,2) is orange (level-2 join). The nodes are connected by horizontal lines. Blue arrows point from (0,2) to (1,2) and from (2,2) to (1,2).

Vertical connections between rows are shown by lines. The vertical axis is labeled $\text{join } c(2)=5$.

Legend:

- L_0 copy
- level-1 join
- level-2 join
- level-3 join

To develop some intuition on what (1) says about a vertex v in L_n^c : the number $|v| - 1$ indicates that v was added that many stages ago in the recursion, at the position of p_k in the path joining the two copies of the previous paths; the sequence t tells the story, in order, of which of the two copies of the previous path the vertex lies in. In other words, vertices of the form $(p_k)^\frown t$ for $t \in 2^n$ are precisely those that were copied from the original L_0^c in all possible positions along L_n^c , doubling the number of copies at each stage.

Based on this discussion, a vertex in some L_n^c is completely determined by the stage $n - m$ at which it was added in the position of p_k and then copied according to the binary sequence t .

3 A notion of smallness for copies

Now, define a notion of smallness analogous to the one presented in [Ber24]. By definition, the set of edges of G is the continuous image of some Polish space E , let us say under π . For a finite graph H , consider $\text{Hom}(H, G)$ as the set of all maps

$$\begin{aligned}\varphi &: V(H) \rightarrow X \\ \varphi &: E(H) \rightarrow E\end{aligned}$$

that satisfy that for all $uv \in E(H)$, $\pi(\varphi(uv)) = \varphi(u)\varphi(v)$.¹ Recall that $X = V(G)$ is the Polish space of vertices on which the analytic graph G is defined. Notice that $\text{Hom}(H, G)$ is also Polish space, being a closed subset of the product space $X^{V(H)} \times E^{E(H)}$.

Given $\mathcal{H} \subseteq \text{Hom}(H, G)$, define $\mathcal{H}(u) := \{\varphi(u) : \varphi \in \mathcal{H}\}$, which is analytic whenever $\mathcal{H}$ is Borel. Also, let $\mathcal{H}(uv) := \{\varphi(uv) : \varphi \in \mathcal{H}\}$.

Definition 3.1.

1. Let $k < \omega$. Say that the set A has property $\Phi(A, k)$ if $A \subseteq X$ is an analytic set of vertices of G such that all odd walks of G with endpoints in A have length at most $2k - 1$.
2. A Borel set $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is called *tiny* if there is a vertex $u \in V(L_n^c)$ and a natural k such that $\Phi(\mathcal{L}(u), k)$.
3. A set $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is *small* if it is the countable union of tiny sets. This notion defines a σ -ideal.
4. A set is *large* if it is not small. Notice that when H is the graph with a single vertex $H = \bullet (= L_0^c)$, $\text{Hom}(H, G)$ can be identified with X and then for any $\mathcal{H} \subseteq \text{Hom}(\bullet, G)$, $\mathcal{H}(\bullet)$ is identified with $\mathcal{H}$.

Now invoke the following auxiliary result that links the Φ property to Borel 2-colourability. The proof proceeds by induction on n using analytic separation; see [CMSV21, Claims 3.3–3.5] for details. Recall that a set of vertices of G is *invariant* if it is the union of connected components of G . In addition, if $A \subseteq X$, then $[A]_{E_G}$ is the union of all connected components of G that intersect A , that is, the set of vertices that can be reached from some element of A by a finite path. Therefore, a set A is invariant if and only if $A = [A]_{E_G}$.

Lemma 3.2 (Claim 3.5 of [CMSV21]). *If $\Phi(A, n)$, then there is an invariant Borel set $B \supseteq [A]_{E_G}$ that induces a 2-Borel-colourable subgraph of G , that is, there is a Borel proper colouring $c_B: G \upharpoonright_B \rightarrow 2$.*

¹Formally, φ is a pair of two maps, however one can assume that $V(H) \cap E(H) = \emptyset$ and then there is no risk of confusion.

Proposition 3.3. *If $\chi_B(G) > 2$ then X (via its identification with $\text{Hom}(\bullet, G)$) is large.*

Proof. Suppose that $\text{Hom}(\bullet, G) = \bigcup_{m < \omega} \mathcal{H}_m$ where each $\mathcal{H}_m$ is a tiny Borel set. For every $m < \omega$, Lemma 3.2 produces an invariant Borel set $B_m \supseteq [\mathcal{H}_m]_{E_G}$ with a Borel 2-colouring c_m . Define $c: X \rightarrow 2$ by letting $m(x) := \min\{m < \omega : x \in B_m\}$ and $c(x) = c_{m(x)}(x)$. Then c is a Borel 2-colouring because each B_m is invariant. $\square$

Given that L_{n+1}^c is obtained from two copies of L_n^c joined by a path, it is convenient to have a means to referring to these distinct copies. Keeping property (1) in mind, for every $\varphi \in \text{Hom}(L_{n+1}^c, G)$ and $i < 2$, define $\varphi^i \in \text{Hom}(L_n^c, G)$ by $\varphi^i(u) = \varphi(u \smallfrown (i))$ for all $u \in V(L_n^c)$ and $\varphi^i(uv) = \varphi(u \smallfrown (i), v \smallfrown (i))$ for all $(u, v) \in E(L_n^c)$. Thus, given $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$, define $\mathcal{L}^{+c}$ as the set of all $\varphi \in \text{Hom}(L_{n+1}^c, G)$ for which $\varphi^i \in \mathcal{L}$ for both $i < 2$. Notice that if $\mathcal{L}$ is Borel, then so is $\mathcal{L}^{+c}$.

The length of the path added needs to be updated dynamically throughout the proof. Formally, one adjusts the parameter c by moving into a different branch of the Baire space when constructing the graphs L_n^c . Notice that, for two $c, d \in \omega^\omega$ and for every $n < \omega$, the condition $d \upharpoonright_n = c \upharpoonright_n$ implies that $L_k^c = L_k^d$ for all $k < n$.

Lemma 3.4.

- (a) *Suppose that $n > 0$, that c only takes odd values, that k is a natural and $t \in 2^{<n}$ is such that $(p_k) \smallfrown t \smallfrown (i)$, for $i < 2$, are vertices of L_n^c (see (1)). Then, they are an odd distance apart in L_n^c .*
- (b) *For any $n < \omega$, if $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is a large Borel set, then for any $N < \omega$ there exists $d \in \omega^\omega$ such that $d \upharpoonright_n = c \upharpoonright_n$, $d(n) \geq N$ is odd, and $\mathcal{L}^{+d}$ is a nonempty subset of $\text{Hom}(L_{n+1}^d, G)$.*
- (c) *For any $n < \omega$, if $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d \upharpoonright_n = c \upharpoonright_n$ and $\mathcal{L}^{+d}$ is a large subset of $\text{Hom}(L_{n+1}^d, G)$.*

Proof. For (a), observe that both vertices come from the same vertex $(p_k) \smallfrown t$ in $L_{|t|+1}^c$, and so their distance in L_n^c is twice the distance from $(p_k) \smallfrown t$ to $e_1^{|t|}$ in $L_{|t|+1}^c$ (since the two vertices are each in distinct copies of $L_{|t|+1}^c$ inside L_n^c and the unique path between them passes through the joining path at stage $|t|$), plus the length of the joining path at that stage, which is $c(|t|) + 2$. Since $c(|t|)$ is odd, this distance is odd.

For (b), since $\mathcal{L}$ is large, it is not tiny. Hence, for every vertex $u \in V(L_n^c)$ and every natural k , $\Phi(\mathcal{L}(u), k)$ fails. In particular, fix the gluing vertex $u_0 := e_1^n \in V(L_n^c)$ (recall that $e_1^0 = p_0$). Then for every k there exist $\varphi_0, \varphi_1 \in \mathcal{L}$ and an odd walk $(v_0, v_1, \dots, v_m)$ in G with $v_0 = \varphi_0(u_0)$, $v_m = \varphi_1(u_0)$, and $m > 2k - 1$. Choosing k large enough, it is ensured that $m \geq N + 2$ and m is odd. Since m is odd, $d(n) := m - 2$ is also odd and satisfies $d(n) \geq N$. For each $1 \leq j \leq m$, since $(v_{j-1}, v_j) \in E(G) = \pi[E]$, pick $e_j \in E$ with $\pi(e_j) = (v_{j-1}, v_j)$. Setting $d(x) = c(x)$ for all $x \neq n$ and noting $L_n^c = L_n^d$ (since $d \upharpoonright_n = c \upharpoonright_n$), define $\varphi \in \text{Hom}(L_{n+1}^d, G)$ by $\varphi^0 = \varphi_0$, $\varphi^1 = \varphi_1$, $\varphi(p_i) = v_{i+1}$ for $0 \leq i \leq m - 2 = d(n)$,

and the edge assignments $\varphi(u_0^\wedge(0), p_0) = e_1$, $\varphi(p_j, p_{j+1}) = e_{j+1}$ for $0 \leq j < m-2$, and $\varphi(p_{m-2}, u_0^\wedge(1)) = e_m$. Then $\varphi \in \mathcal{L}^{+d}$.

Next, prove (c) by contradiction. Recall that constructing L_{n+1}^c from L_n^c only requires knowing the value of $c(n)$. Suppose that for all d that agree with c except possibly in the n -th value, $\mathcal{L}^{+d} = \bigcup_m \mathcal{F}_m^d$ where each $\mathcal{F}_m^d$ is a tiny Borel subset of $\text{Hom}(L_{n+1}^d, G)$. For each m and d , let $w_m^d \in V(L_{n+1}^d)$ and $k_m^d < \omega$ be such that $\Phi(\mathcal{F}_m^d(w_m^d), k_m^d)$.

Since $w_m^d \in V(L_{n+1}^d)$, it is either a *non-path vertex* of the form $u^\wedge(i)$ for some $u \in V(L_n^c)$ and $i < 2$, or a *path vertex* p_j for some $0 \leq j \leq d(n)$ in the joining path. We claim that in either case, there exists a vertex $u_m^d \in V(L_n^c)$ and $\ell_m^d < \omega$ such that $\Phi(\mathcal{F}_m^d(u_m^d \wedge (i_m^d)), \ell_m^d)$ for some $i_m^d < 2$ where $u_m^d \wedge (i_m^d)$ is a non-path vertex of L_{n+1}^d .

Indeed, if $w_m^d = u^\wedge(i)$ is already non-path, take $u_m^d = u$, $i_m^d = i$, $\ell_m^d = k_m^d$. If $w_m^d = p_j$ is a path vertex of L_{n+1}^d , then the joining path connects $e_1^{n \wedge}(0)$ to $e_1^{n \wedge}(1)$ via $p_0, \dots, p_{d(n)}$. Since p_j and $e_1^{n \wedge}(0)$ are at distance $j+1$ in L_{n+1}^d , every homomorphism $\varphi \in \mathcal{F}_m^d$ maps them to vertices connected by a walk of length $j+1$ in G . Given any odd walk $(a_0, \dots, a_r)$ in G with $a_0, a_r \in \mathcal{F}_m^d(e_1^{n \wedge}(0))$, witnessed by $\varphi, \varphi' \in \mathcal{F}_m^d$, the walk $(\varphi(p_j), \dots, \varphi(e_1^{n \wedge}(0))) = a_0, \dots, a_r = \varphi'(e_1^{n \wedge}(0)), \dots, \varphi'(p_j)$ has length $r+2(j+1)$, which has the same parity as r (hence is also odd), and has endpoints in $\mathcal{F}_m^d(p_j)$. By $\Phi(\mathcal{F}_m^d(p_j), k_m^d)$, $r+2(j+1) \leq 2k_m^d - 1$, so $r \leq 2(k_m^d - j - 1) - 1$. It follows that $\Phi(\mathcal{F}_m^d(e_1^{n \wedge}(0)), \max\{0, k_m^d - j - 1\})$. Thus, set $u_m^d = e_1^n$, $i_m^d = 0$, $\ell_m^d = \max\{0, k_m^d - j - 1\}$.

Claim 3.4.1. For each m and d , there is a Borel set $B_m^d \supseteq \mathcal{F}_m^d(u_m^d \wedge (i_m^d))$ such that $\Phi(B_m^d, \ell_m^d)$.

Proof of Claim. Use the First Reflection Theorem (Lemma 1.4). Fix $\ell = \ell_m^d$. It suffices to verify that $\Phi_\ell := \{A \subseteq X : \Phi(A, \ell)\}$ is Π_1^1 on Σ_1^1 . Let Y be Polish and $A \subseteq Y \times X$ be Σ_1^1 . Fix a Polish space N and a continuous surjection $g: N \rightarrow A$. For each odd $r \geq 2\ell + 1$, define

$$C_r := \{(y, x_0, \dots, x_r, n_0, n_1, \hat{e}_1, \dots, \hat{e}_r) \in Y \times X^{r+1} \times N^2 \times E^r : \\ g(n_0) = (y, x_0) \wedge g(n_1) = (y, x_r) \wedge \bigwedge_{j < r} \pi(\hat{e}_{j+1}) = (x_j, x_{j+1})\}.$$

Then C_r is closed (by the continuity of g and π), and $D_r := \text{proj}_Y(C_r)$ is Σ_1^1 . Hence $D := \bigcup\{D_r : r \text{ odd}, r \geq 2\ell + 1\}$ is Σ_1^1 , and $A_{\Phi_\ell} = \{y \in Y : A_y \in \Phi_\ell\} = Y \setminus D$ is Π_1^1 . Since $\mathcal{F}_m^d(u_m^d \wedge (i_m^d))$ is analytic and satisfies $\Phi(\cdot, \ell)$, the First Reflection Theorem yields the desired Borel. $\dashv$

Take

$$\mathcal{H}_m^d = \{\varphi \in \mathcal{L} : \varphi(u_m^d) \in B_m^d\},$$

which is also a Borel subset of $\text{Hom}(L_n^c, G)$. We claim that $\mathcal{H}_m^d$ is tiny. Indeed, $\mathcal{H}_m^d(u_m^d) = \{\varphi(u_m^d) : \varphi \in \mathcal{L} \wedge \varphi(u_m^d) \in B_m^d\} \subseteq B_m^d$, so $\Phi(\mathcal{H}_m^d(u_m^d), \ell_m^d)$.

Now, the set of pairs (u_m^d, i_m^d) ranges over the finite set $V(L_n^c) \times 2$ (which does not depend on d), and (m, d) range over countably many values. Thus $\bigcup_{m,d} \mathcal{H}_m^d$ is a countable

union of tiny sets, hence small. The set $\mathcal{L}_- := \mathcal{L} \setminus \bigcup_{m,d} \mathcal{H}_m^d$ is therefore large (and Borel). By part (b), there exist φ and d such that $\varphi \in \mathcal{L}_-^{+d} \subseteq \mathcal{L}^{+d}$. Then $\varphi \in \mathcal{F}_m^d$ for some m . It follows that $\varphi^{i_m^d}(u_m^d) = \varphi(u_m^d \wedge (i_m^d)) \in \mathcal{F}_m^d(u_m^d \wedge (i_m^d)) \subseteq B_m^d$, hence $\varphi^{i_m^d} \in \mathcal{H}_m^d$. But $\varphi \in \mathcal{L}_-^{+d}$ implies $\varphi^{i_m^d} \in \mathcal{L}_-$, contradicting $\mathcal{L}_- \cap \mathcal{H}_m^d = \emptyset$. $\square$

4 Construction of a minimal graph of Borel chromatic number at least 3

The goal of this section is to construct a family of graphs $\mathbb{L}_c$ indexed by $c \in \omega^\omega$.

Fix c and consider X_c as the set of all tuples $(m, k, x) \in \mathbb{N} \times \mathbb{N} \times 2^\mathbb{N}$ such that either $m = 0$ and $k = 0$, or $m \geq 1$ and $k \leq c(m-1)$; this is a closed subspace of the product space, and hence Polish. Define, for each (m, k, x) with $m \leq n$, $\pi_n(m, k, x)$ as the vertex of L_n^c determined by these parameters. Formally, $\pi_n(m, k, x) := (p_k)^\wedge x \upharpoonright_{n-m}$. For example, $\pi_3(1, 2, 0110 \cdots) = (p_2, 0)$ can be seen in Figure 1 labeled as $(2, 0)$.

Finally, $\mathbb{L}_c$ is the graph on X_c where two (n_i, k_i, x_i) , for $i < 2$, are adjacent if and only if the $\pi_n(n_i, k_i, x_i)$ are adjacent in all L_n^c with $n \geq \max\{n_0, n_1\}$. Some basic properties of $\mathbb{L}_c$ follow.

Lemma 4.1.

- (a) Two vertices $(n_i, k_i, x_i) \in X_c$, for $i < 2$, are in the same connected component of $\mathbb{L}_c$ if and only if there are $t_i \in 2^{<\omega}$ and $x \in 2^\omega$ such that $|t_0| - |t_1| = n_1 - n_0$ and $x_i = t_i^\wedge x$.
- (b) $\mathbb{L}_c$ has no cycles and all vertices of $\mathbb{L}_c$ have degree 2 except for the vertex $(0, 0, \{0\}^\omega)$, which has degree 1.
- (c) If, additionally, c takes only odd values, then $\chi_B(\mathbb{L}_c) = 3$.

Proof. For (a), let $(n_i, k_i, x_i) \in X_c$ for $i < 2$. If they are in the same connected component, then for all large enough n , $\pi_n(n_0, k_0, x_0)$ and $\pi_n(n_1, k_1, x_1)$ lie in the same copy of L_{n-1}^c inside L_n^c . By the construction of L_n^c , two vertices $(p_{k_0})^\wedge x_0 \upharpoonright_{n-n_0}$ and $(p_{k_1})^\wedge x_1 \upharpoonright_{n-n_1}$ are in the same copy of L_{n-1}^c in L_n^c if and only if their last binary digits agree, that is, $x_0(n-n_0-1) = x_1(n-n_1-1)$ for all large n . Writing $x_0 = t_0^\wedge x$ and $x_1 = t_1^\wedge x$ with $|t_0| = n_1 - n_0 + |t_1|$ (assuming $n_0 \leq n_1$), one obtains the stated condition. Conversely, if $x_i = t_i^\wedge x$ with $|t_0| - |t_1| = n_1 - n_0$, then for all large n , $\pi_n(n_0, k_0, x_0) = (p_{k_0})^\wedge t_0^\wedge x \upharpoonright_r$ and $\pi_n(n_1, k_1, x_1) = (p_{k_1})^\wedge t_1^\wedge x \upharpoonright_r$ for some r , so they share a common tail and lie in the same connected component of L_n^c , hence of $\mathbb{L}_c$.

For (b), every vertex of L_n^c has degree at most 2, so the same holds for $\mathbb{L}_c$. It follows from acyclicity of the finite paths L_n^c that $\mathbb{L}_c$ is acyclic. The vertex $(0, 0, 0^\omega)$ maps under π_n to $e_0^n = (p_0)^\wedge 0^n$, which is an endpoint of L_n^c , hence has degree 1 in L_n^c for all $n \geq 1$. Thus

$(0, 0, 0^\omega)$ has degree 1 in $\mathbb{L}_c$. For any other vertex $(m, k, x) \in X_c$, the image $\pi_n(m, k, x)$ is an interior vertex of L_n^c for all sufficiently large n , so it has degree exactly 2.

(c) That $\chi_B(\mathbb{L}_c) \leq 3$ follows from Proposition 1.2 and part (b). Argue the other inequality by contradiction. Suppose that there is a Borel partition of X_c into two independent sets $X_c = I_0 \cup I_1$. Since X_c is Polish, the Baire category theorem guarantees that some I_ε is non-meager, hence comeager in a basic open set $U = \{(m, k, x) \in X_c : x \upharpoonright_r = t\}$ for some m, k, r , and $t \in 2^r$. In particular, for a comeager set of $x \in 2^\mathbb{N}$ extending t , $(m, k, x) \in I_\varepsilon$. Since $(m, k, t^\frown(0)^\frown y)$ and $(m, k, t^\frown(1)^\frown y)$ are in the same connected component (by part (a)) for any $y \in 2^\mathbb{N}$, and I_ε is comeager in U , one can find y such that both $(m, k, t^\frown(0)^\frown y)$ and $(m, k, t^\frown(1)^\frown y)$ lie in I_ε . But by Lemma 3.4(a), these two vertices are an odd distance apart in $\mathbb{L}_c$ (as $(p_k)^\frown t^\frown(0)$ and $(p_k)^\frown t^\frown(1)$ are an odd distance apart in every L_n^c for large enough n), contradicting I_ε being independent. $\square$

The main result is then split into two parts.

Theorem 4.2. *For any analytic graph G , either $\chi_B(G) \leq 2$ or there is a c such that $\mathbb{L}_c \rightarrow_c G$.*

Moreover, c can be chosen to be unbounded and take only odd values.

Theorem 4.3. *Let $c, d \in \omega^\omega$ be unbounded and only taking odd values, then $\mathbb{L}_c$ and $\mathbb{L}_d$ are continuously homomorphically equivalent.*

The proof of Theorem 4.3 is purely combinatorial: one constructs homomorphisms between $\mathbb{L}_c$ and $\mathbb{L}_d$ by iteratively extending partial maps along finite path components, using the “large gap property” guaranteed by unboundedness. The argument does not depend on the method used to establish Theorem 4.2; see [CMSV21, Proposition 4.2 and Claim 4.1] for details.

Combining Theorems 4.2 and 4.3, and fixing any unbounded odd c_0 (e.g., $c_0(0) = 1$ and $c_0(n) = 2n - 1$ for $n > 0$)², the L_0 dichotomy of [CMSV21] is recovered:

Corollary 4.4. *Let $\mathbb{L}_0 := \mathbb{L}_{c_0}$. For any analytic graph G on a Polish space, exactly one of the following holds:*

- (a) $\chi_B(G) \leq 2$;
- (b) $\mathbb{L}_0 \rightarrow_c G$.

5 Proof of Theorem 4.2

By Lemma 4.1(c) and Fact 1.1, it is clear that both conditions cannot hold simultaneously. Now suppose that G is an analytic graph of Borel chromatic number at least three. The theorem is proved by constructing a valid c and a continuous homomorphism witnessing $\mathbb{L}_c \rightarrow_c G$.

²It is perhaps worth noting that I picked a *different* parameter for Figure 1 purely for aesthetic purposes.

By Proposition 3.3, X is large. Fix compatible complete metrics ρ on X and δ on E . Since $L_0^c = \bullet$, $\text{Hom}(L_0^c, G)$ may be identified with X and set $\mathcal{L}_0 := X$, which is large. Recursively construct a function $c \in \omega^\omega$ (taking only odd values) and a sequence of large Borel sets $\mathcal{L}_n \subseteq \text{Hom}(L_n^c, G)$ such that for all n ,

- (1) $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$;
- (2) for all $u \in V(L_n^c)$, $\text{diam}_\rho(\overline{\mathcal{L}_n(u)}) < 2^{-n}$; and
- (3) for all $(u, v) \in E(L_n^c)$, $\text{diam}_\delta(\overline{\mathcal{L}_n(u, v)}) < 2^{-n}$.

Given $\mathcal{L}_n$ large, apply Lemma 3.4(c) to obtain $d \in \omega^\omega$ with $d \upharpoonright_n = c \upharpoonright_n$ and $\mathcal{L}_n^{+d}$ large; set $c(n) := d(n)$. To arrange conditions (2) and (3) for $n+1$: since X and E can each be covered by countably many closed sets of ρ -diameter (respectively δ -diameter) less than $2^{-(n+1)}$, and small sets form a σ -ideal, one may intersect with the preimages of these covers and discard the resulting small pieces, passing to a large Borel subset $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$ satisfying (2) and (3) at stage $n+1$. Moreover, by choosing $c(n)$ sufficiently large using part (b) at each stage, c can be made unbounded.

By (1), for all $n \geq m$,

$$\mathcal{L}_{n+1}(\pi_{n+1}(m, k, x)) \subseteq \mathcal{L}_n(\pi_n(m, k, x));$$

indeed, given that $\pi_{n+1}(m, k, x) = \pi_n(m, k, x) \frown (x(n-m))$, each $\varphi \in \mathcal{L}_{n+1}$ satisfies $\varphi^{x(n-m)} \in \mathcal{L}_n$ and

$$\varphi^{x(n-m)}(\pi_n(m, k, x)) = \varphi(\pi_{n+1}(m, k, x)).$$

By the same reasoning, if $\pi_n(m_0, k_0, x_0)$ and $\pi_n(m_1, k_1, x_1)$ are adjacent in L_n^c for all large n , then

$$\begin{aligned} & \mathcal{L}_{n+1}(\pi_{n+1}(m_0, k_0, x_0), \pi_{n+1}(m_1, k_1, x_1)) \\ & \subseteq \mathcal{L}_n(\pi_n(m_0, k_0, x_0), \pi_n(m_1, k_1, x_1)). \end{aligned}$$

Thus, the sets $\overline{\mathcal{L}_n(\pi_n(m, k, x))}$ are closed and nested, with diameters tending to zero (see (2) from before). Since ρ is a complete metric, Cantor's intersection theorem applies. For $(m, k, x) \in X_c$, let $f(m, k, x)$ be the unique point in

$$\bigcap_{n=m}^{\infty} \overline{\mathcal{L}_n(\pi_n(m, k, x))} = \bigcap_{n=0}^{\infty} \overline{\mathcal{L}_{n+m}((p_k) \frown x \upharpoonright_n)}.$$

The map $f: X_c \rightarrow X$ is continuous.

It remains to check that it is a homomorphism from $\mathbb{L}_c$ to G . Suppose that $(m_i, k_i, x_i)_{i < 2}$ are adjacent in $\mathbb{L}_c$, that is, for all $n \geq n_0 := \max\{m_0, m_1\}$, $\pi_n(m_i, k_i, x_i)$ are adjacent in L_n^c .

Similar to the preceding paragraph, the fact that δ is a complete metric implies that there is a unique edge $e \in E$ inside

$$\bigcap_{n \geq n_0} \overline{\mathcal{L}_n(\pi_n(m_0, k_0, x_0), \pi_n(m_1, k_1, x_1))}.$$

By the continuity of π , $(f(m_0, k_0, x_0), f(m_1, k_1, x_1)) = \pi(e) \in E(G)$, in other words, the images form an edge of G , and this concludes the proof.

The hereby presented argument shows that the core of the L_0 dichotomy, namely the construction of a continuous homomorphism from an $\mathbb{L}_c$ graph into any analytic graph of Borel chromatic number at least three, admits a proof that closely parallels Bernshteyn's argument for the G_0 dichotomy, with the challenge of adding a dynamically-changing parameter to the construction.

The main new ingredient is the replacement of the independence-based σ (sets whose vertex projections are independent) with the odd-walk-bounded σ -ideal defined by the Φ property. It remains an interesting question whether analogous adaptations can yield simplified proofs of other dichotomy results in Borel combinatorics, or even entirely new ones.

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